Question ID: 03503d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is $\$14,000$, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

Answer

A. $\$148.15$

B. $\$161.53$

C. $\$172.84$

D. $\$184.94$

Correct Answer: B

Rationale

Choice B is correct. It's given that a business owner plans to purchase 81 chairs. If p is the price per chair, the total price of purchasing 81 chairs is $81p$. It's also given that 7% sales tax is included, which is equivalent to $81p$ multiplied by 1.07, or $81(1.07)p$. Since the total budget is $\$14,000$, the inequality representing the situation is given by $81(1.07)p \leq 14,000$. Dividing both sides of this inequality by $81(1.07)$ and rounding the result to two decimal places gives $p \leq 161.53$. To not exceed the budget, the maximum possible price per chair is $\$161.53$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the maximum possible price per chair including sales tax, not the maximum possible price per chair before sales tax.

Choice D is incorrect. This is the maximum possible price if the sales tax is added to the total budget, not the maximum possible price per chair before sales tax.